

ARCHIVAL RESULT NOTE

DOC-2-019 R0 | Science Program

Short-Range Markov Sequence-Surprise Features vs. Column-Shuffled RNA Across Held-Out Rfam Families

*This is an archival result note, not a research paper and not a reproducible publication package.
It records only verified aggregate facts of the result.*

Status: `publication_blocked` = `artifact_loss` | `reproducibility_package` = `unavailable`

1. Verified Aggregate Facts

The following facts are recorded exactly as verified for the locked R0 run. No other numeric detail is asserted by this note, and none should be reconstructed from secondary summaries.

Item	Verified value
Result identifier	DOC-2-019 R0
Question tested	Short-range Markov sequence-surprise features detect authentic vs. column-shuffled RNA across completely held-out Rfam families
Corpus	2,110 Rfam families / 15,658 authentic seed sequences
Held-out test partition	427 families, completely held out from training
Primary statistic	Family-macro AUROC 0.7043
Uncertainty	95% family-bootstrap CI 0.6835-0.7258
Integrity gates	Data, nuisance, and length-stratum gates PASSED
Performance gates	Locked AUROC ≥ 0.75 FAILED; locked CI lower ≥ 0.70 FAILED
Verdict	Informative negative
Protocol lock SHA-256	778a9fcc967a1d4a2d6594bae8629e21ec839f183770eb9d057a50fb050cd6f8
Data source	Rfam seed alignments (source-dependent)

Recorded 2026-09-22 from the orchestrator's verified summary of the R0 run.

2. Claim Boundary of This Note

The R0 run did not test any named RNA model or generator. The result cannot support a universal RNA hallucination-detector claim, and it admits no post-hoc threshold tuning: the locked AUROC and CI bars are the only criteria under which a positive claim was permitted, and both failed while all integrity gates passed. This note asserts nothing beyond the aggregate facts in Section 1.

3. Artifact-Loss Statement

The original R0 artifacts - the locked protocol text, run code, gate-by-gate results files, and per-family predictions - are not currently recoverable from the stores available to this program. Full reproducibility publication of R0 is therefore BLOCKED (publication_blocked = artifact_loss; reproducibility_package = unavailable).

No evidence placeholder has been filled from transcript summaries or other secondary text. Transcript text is not a substitute for artifacts and has not been transformed into

tables, hashes, rerun claims, or any other surrogate evidence. The styled full paper and technical report scaffolds prepared for this result are preserved but remain UNPUBLISHED unless and until the original R0 bytes are recovered from an original attachment store.

4. Continuation

The properly sealed continuation of this line of work is result R1.1. Pointer as recorded by the orchestrator:

Item	Value
Continuation result	R1.1 (properly sealed)
Repository commit	67d7effc8c4915a27c3f69b5cb00a33d8c0eeb09
Drive copy	Verified copy exists in the program Drive folder
Archive SHA	4521bfc...a33 (as recorded, truncated)

R1.1, not this note, is the authoritative sealed artifact for the lineage. This note exists so that the R0 aggregate result remains citable and the reason for its non-publication is unambiguous.

5. Sealing Record

This note is sealed in two copies: (i) committed to the science-program repository (via the program coordinator), and (ii) placed in the program Google Drive folder. SHA-256 digests of both file formats of this note are reported alongside the sealed copies and reproduced below.

SHA-256 digests of the DOCX and PDF renderings of this note are recorded in the external file MANIFEST.sha256, sealed alongside both copies (a document cannot contain its own final digest). Verification: sha256sum -c MANIFEST.sha256.

End of archival result note.